

Project Proposal: World Happiness Report Analysis and Machine Learning Prediction

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June 1, 2026

1 Introduction

The purpose of this project is to analyze the World Happiness Report dataset and investigate the factors that influence happiness scores across countries. In addition, machine learning techniques will be used to predict happiness scores based on socioeconomic indicators.

2 Dataset

The dataset is based on the World Happiness Report from 2015 to 2019.

The combined dataset contains information about countries and their happiness scores, along with several explanatory variables:

- GDP per capita (economic condition)
- Social support (social relationships)
- Healthy life expectancy (health condition)
- Freedom to make life choices (personal freedom)
- Generosity (charitable behavior)
- Perceptions of corruption (trust in institutions)

The datasets from multiple years were merged into a single CSV file for analysis.

3 Proposed Analysis

The following analyses will be performed:

- Compare happiness scores among countries

- Identify factors most strongly related to happiness
- Examine trends across different years
- Analyze correlations between variables

4 Proposed Visualizations

The project will include the following visualizations:

- Bar charts of happiest countries
- Scatter plots of GDP vs Happiness Score
- Scatter plots of Social Support vs Happiness Score
- Correlation heatmap
- Yearly happiness trends

5 Machine Learning Approach

Machine learning regression models will be used to predict Happiness Score.

Input features:

- GDP per capita
- Social support
- Healthy life expectancy
- Freedom to make life choices
- Generosity
- Perceptions of corruption

Target variable:

- Happiness Score

The initial model will use Linear Regression. Additional models such as Random Forest Regression may also be explored.

6 Expected Outcome

This project is expected to identify important factors affecting happiness and demonstrate how machine learning can predict happiness scores using socioeconomic indicators.

7 References

- World Happiness Report Dataset (2015–2019), Kaggle.
- Helliwell, J., Layard, R., Sachs, J. World Happiness Report.